

Octopus: “Cognitive Bifurcations and Mathematical Explorations”

Source: [YouTube](#) Playlist: 4th Applied Active Inference Symposium (2024)

Duration: 37:43 | Uploaded: Unknown Transcript method: auto_caption

abstract for this talk was and I had to look this up this morning uh thinkers and thoughts have a co-evolutionary relationship this relationship is especially clear in the arena of mathematics where mathematicians forage for mathematical statements and structures in a process governed by complex and potentially unstable rules we will give a partial mapping of the structure of this space which necessarily relies not only on rigorous mathematical logic but also on subject beliefs metacognitive juristic and creative play so I've done a magic trick here all of these things are claims that are in need of evidence from my perspective and the goal here will be to provide some evidence in that direction but also to engage in conversation so in the construction of expertise I have no ambition here to play the role of expert there are certainly qualifications and evidence that I could point to my experience as a math professor for seven and half years my lived experience as a neurodivergent individual for 42 years my time spent in an octopus identity for the last three years openly and none of those things makes what I say here more valuable than what you think about them the only request I will make is that I've given a lot of effort into thinking through these thoughts some of which were quite difficult that showed up for me and I ask that you will give some level of effort towards finding the value in those thoughts and how they can be mapped to your reality so let's move from the abstract to abstraction when people think of mathematics in the modern sense they often begin here as both its value and its difficulty we might imagine that Bob and Alice both have some cows Bob has two and Alice has three and they're curious how many cows they would have if they combine their herd Alice can bring her cows to Bob and they can count them then if they have the physical process of counting at their disposal a cognitive tool of one-to-one mapping between objects and objects but they can also do it conceptually one of the things that abstraction enables is the ability to move information about physical perceptions more quickly than the physical things themselves can move in other words Alice doesn't have to bring her cows with her if she can bring the concept of three or the idea that she has three so

this is another aspect of mathematics that is often highly valued and highly contentious in relation to the ways in which we realize and reify scientific practice education and other ways in which education other ways in which mathematics shows up in our lives is how are we actually giving a physical body to the concept that we are hoping to represent the fact that two apples are conceptually in some sense the same as two cows or two fingers is one level a level above that is the fact that we can attribute symbols to these representations anything from two interlaced eyes and the Roman numeral setting to two in the Hindu Arabic numeral setting to words like two and dots to two fingers the two fingers are physical yet somehow they're also a basis of abstraction because the idea of abstraction includes the fact that there must be something concrete to abstract and this two fingers I've included on the page of abstract representations partially because I like the idea of things having bodies I like the idea of the number two having lots of different bodies how many bodies can one idea have but also I like the ambiguity here between the two and the three because in some systems the way that I'm showing this here would be interpreted as three and in others as two so with that ambiguity in mind there is no slide three please do not discuss slide three later we will not bring it up again and in fact it is not advantageous to you to do so because the claim that there is no slide three is an idea wholly owned by eight arms nine brains and if you want to discuss the fact that there is no slide three there will be a licensing fee involved thankfully some ideas are open to discussion with no financial transaction involved and one of them as we finally come to slide four is a very important Concept in mathematical pedagogy called the rule of four I think this encapsulates thinking that might have been going on for a while before it was formally published but the first time I encountered it was in the excellent calculus textbook by Hughes Howen gleon and McCullum at all this was the first calculus textbook I looked at When I Was An undergraduate student and it was also the first calculus textbook I looked at when I was a beginning Professor beginning to teach my first Calculus class I'm afraid that far too many students probably completely skip this section uh which is in the pre Preamble uh but one of the key Concepts that it elucidates is how to think about mathematics and it's not saying this is the right way it's saying this is a way and this is a way that the textbook uses quite profitably it's called the rule of four and it says essentially that when you encounter a new mathematical concept there are four distinct but interrelated ways that one might think about it you could give a verbal description or a definition um you could draw a graph or some type of diagram you could use numbers in the sense of data or you you could use a symbolic representation like a formula or an equation and the extent to which these are distinct to inter

related highlights the value that each of these may have to individuals as a latching on point for thinking about difficult mathematical Concepts and calculus it also eliminates some of the privilege that might exist around specific representations around the idea that symbols might be real math whereas when we explain them at a high level somehow that isn't real math for instance if we look at the types of mathematical works that mathematicians actually produce um it might be surprising to undergraduate students what percentage of this is not what they would traditionally think of as symbolic and what I would mean here is equations and formulas like we get to page three and finally we see something that looks like a symbolic mathematical expression but the majority of the work here is spent on words words and diagrams visuals a large portion of this is in the theorems of course which are something that is uh a very individual feature of pure mathematics but let's not get bogged down there let's simply say there are many different representations and one of the challenges of an effective instructor is to map between these different representations so that people who might feel most at home in one of them can use that as a leverage point for accessing the other one uh based on my background I'm I'm a defro math professor voluntarily also a chaos magician and so I'm very very tempted to try to build a funter between this rule of four originating in mathematics and this four quadrants model of Human Experience Advanced by Ramsey Dukes in his book known as s sopy um which says that there are these four lenses through which human beings can process and structure experience there's art religion magic and science it's so tempting to try to do something like say the symbols are the magic and the words are the religion and the numbers are the science and the graph is the art and I'm going to resist that Temptation because I think it's it might be a useful initial temptation but the resistance of the Temptation provides its value instead I'm going to hold this as an idea we will return to it in a moment where it will play against some other Concepts that we're likely to encounter I will stop here for a moment and say something in promptu though which is um there's a practice in this world of chaos magic called belief shifting where the magician consciously and intentionally adopts a specific belief as if it were something they believe to be true and there's a very related practice in mathematics which is the construction of the axioms or the adoption of the axioms so in mathematical proofs once you adopt specific axioms or specific assumptions the conclusions should follow not without effort but in a clear logical flow mathematicians are capable of adopting different axioms and this produces what I'm going to call a cognitive bifurcation so a cognitive bifurcation since we've already split things into four they may be too much let's try to get things back together and just split it into two one place that a cognitive

bifurcation might show up in the history of mathematics is in the different types of geometry that geometers think about so one of the features of modern geometry is the realization that the parallel postulate in Euclidean Geometry is not possible to prove from the other axioms of Euclidean geometry and so the parallel postulate says that given a line and a point not on that line there is exactly one parallel line or there's actually one line that passes through that point parallel to the original line and if you choose not to adopt this belief as I played with in the idea of one of his lectures um it opens up a whole new world of non-Euclidean geometry both spherical and hyperbolic so the axioms that we adopt and the beliefs that we intentionally choose to consider are powerful tools that can open up new conceptual playgrounds but also quite dangerous Alice and Bob probably feel very safe in this world even if we don't admit a slide three we admit a number three to describe how many cows one of them may have and then the question is are these all the numbers and if they are not in which directions would we be willing to consider perhaps we asking what comes after five or after the number after five and like a child we realize that we can continually add one and there is no a priori upper bound and we come across the idea of infinity perhaps we start to look for what could be in between the numberers and so we discover numbers like one and a half or 1.1 or perhaps we make a dangerous leap in the other direction and we admit the possibility that there is a number which says that you have no cows and while you might need such a number is an intriguing question but let's say that we do have one when we introduce a number in this way we need not only to consider the fact that the number exists we need to consider the ways in which it will interact with the other numbers that exist in other words the rules so as we begin to if we want to we know how to add numbers that were one through five but how do we add to zero and this is a cognitive bifurcation because now we have a symbol that plays by its own rules and we've opened up a whole new world of numbers to where when we added positive numbers together they always increased we were always heading to the right the fact that we can now introduce a number where we add it and identity for the binary operation of addition is uh an interesting an interesting split that opens up more possibilities on the left hand side and I would say this is a cognitive bifurcation it says in the the definition that I'm adapting from Wikipedia a bifurcation occurs when a small smooth change made to the parameter values of a system causes a sudden qualitative or topological change in its Behavior it's hard to imagine a change smaller or smoother than putting the circle here and declaring that it doesn't do anything on addition and yet this an immense cultural controversy when these numbers are being introduced so I'm going to make the claim that mathematical thinking systems are complex

systems I'm going to give the highly unsatisfactory proof by Authority that this claim is obvious one piece of evidence we could point to is that the existence of a parameter bifurcation almost immediately suggest that we're dealing with a complex system and yet that is not quite the right way to think about it especially if we're going to be looking at it from the active inference context so let's ask for more evidence and thus not accept proof by Authority what evidence do we see for the belief that mathematical thinking systems are complex systems and one way that we might uh conceptualize this through a model is at the agent level where we think about the types of agents who are involved in these systems I've heard that there would be people here who like ants so I've made sure to include some and uh here's my aunt Hill and here are my ants and I have these ants foraging for formulas that they bring back to the colony which then feed other ants which are a part of the ecosystem the formulas and the symbols that they bring back in this mathematical world of semantics is is more than just symbols right it's something that actually contributes to the survival of the colony but in order to do that it must be converted into other things so so it would be easy to just try to be efficient through this and reduce it as much as we could to something useful I'm sympathetic to that Viewpoint because mathematics has to be useful at some level to human existence to justify its existence I believe so if we were looking at this on a belief level on a on a probability level we might say something like with certain axioms as priors mathematical theorems are simply beliefs such that an agent in order to survive who quickly will quickly adopt correct beliefs with probability one if you want to make $2 + 3$ equals 6 you will quickly run into issues in which everything collapses and it ceases to be a useful tool and and I'm not opposed to this model I just want to say it's oversimplified because many of the beliefs that mathematicians choose to adopt and choose to spend their valuable resources thinking about don't have any immediate mapping to applications for instance the question of whether or not zero is a natural number is something mathematicians will argue about what is the utility in foraging for this belief perhaps you can think of one um but I think it speaks to something else that could be happening that mathematicians throughout the centuries have seen which is that the the claims of truth that you make are not entirely limited by the axioms that you assume and by the process of logic there's there are larger questions here cultural questions uh of identity ontological questions of what is real what does it mean for zero being a natural number to be a true statement mathematicians find it useful to disagree about this the conversations that we have about these things make us better mathematicians I feel I'll never forget when my PhD adviser explained to me why he believed that zero was a natural number but there's no

mathematical proof that can provide this it's an axiom that you accept or do not when you define these numbers and when we talk about types of numbers there's a famous cognitive vacation that occurred at some point in history which is the division uh the discovery of the fact that not all numbers are rationals not all numbers can be expressed as a ratio between two integers if you're not willing to accept this then you cannot come up with a solution to $x^2 - 2 = 0$ and of course you could by authoritarian fiat simply declare that there is no solution to $x^2 - 2 = 0$ there's a legend that the mathematician hippasus who discovered that the square root of two was irrational um was thrown overboard by the pythagoreans so that he would not share this information and I am imagine this as a kind of cyberpunk reimagining of ancient Greek geometry not as an insult in any way to Pythagoras who I think would appreciate this um but to imagine that there is a geometer who is not opposed to the idea but opposed to the pathway by which it came through and willing to use Force to ensure that that dangerous doorway stays closed so another model of math person perhaps that we could view is as a boundary between wherever math comes from and the quote unquote real world and the real world is in full of Agents already that we can easily map to there are institutions educational institutions governmental institutions um professional societies there's politics where the government may decide certain levels of funding for those institutions business which may choose to fund the institutions to uh insert itself into the process of politics there are funding sources there's media one thing that occurred to me as I Was preparing this talk is we don't often think of mathematical journals or scientific journals as media but they are they're media that's made and constructed like any other and then is the question of where math comes from where does math come from uh this is what I've wrestled with a while one reason is it opens up some very some doors that from the material materialist utilitarian perspective of like a School of Engineering feel really strange where did Alice and Bob first get that idea of numbers and does doing math involve some type of contact with these strange things that could happen we don't see very much of this in our schooling which is often about producing symbols uh producing people who can follow precise rules with respect to symbolic manipulation and at that point it can become a type of overcorrection and I'm sympathetic to this door that uh I'm sympathetic to this perspective that opens the door a bit more to the strange not only by my own temperament but also through the lived experience of people who accomplished very effective physical and Engineering uses of mathematics but were willing to express that they were sympathetic to this viewpoint so Hinrich Herz For Whom the the unit of a cyclic repetition and a computer processing unit is named for says one cannot escape the

feeling that these mathematical formulas have an independent existence and in intelligence of their own that they are wiser than we are wiser even than their discoverers in other words if the intelligence is in the symbols and not in the humans who are trying to work with them as tools this provides a very different perspective on the creation of mathematical thought and on the ways in which mathematical thought not only originates but also can be applied it also points out to some extent the horror that we can be doing when we do this non-consensual cognitive bifurcation of a person of people persons into math persons and not math persons Often by their ability to match some preconceived rule to arrive at a correct answer often with an overemphasis on speed so as we think about the fact that the people the persons the mathematicians the math persons may be foraging for these symbols and formulas to bring back to the real world it's what happens if we begin to give if we as hers it begin to give these symbols their own ontology and their own agency to realize that they are also likely out in the wild searching for the people to whom with whom they will engage productively so this takes us back to our stranger ideas where we intentionally left the door open and here I'm indebted to Gard mayor who in his article magicians of the 21st century um pointed to Ramsey dukes's for quadrants model as splitting reality into the irrational and the rational intentionally left the rational box off here because I wasn't sure where to put it I would put religion in the rational category because there are very clear apologetic logical arguments it's clear that religion at least attempts to play in the same sphere of the rational as science does whereas art and Magic would be in the irrational category and I am intentionally echoing the splitting here of the real numbers into rational irrational with whatever consequences this might have the quote hinr Hertz again outside our Consciousness there lies the cold and alien world of actual things between the two stretches the narrow Borderland of the senses no communication between the two worlds is possible excepting across the narrow strip for a proper understanding of ourselves and of the world it is of the highest importance that this Borderland should be thoroughly explored and the problem as map makers and an Ever evolving and somewh say chaotic territory certainly highly Dynamic of this moment and SpaceTime where we find ourselves is where to draw where is the border if this is the border that we're supposed to be exploring for a proper understanding of ourselves and of the world how do we know when we are on the border and when we may be inadvertently Crossing it and how can we enrich our tools for finding each side and productively bringing ideas from one side to the other or whatever Travelers are wishing to cross whatever we call them I've late recently been reading this book symbolism by Alfred North Whitehead and he speaks to this but he speaks to it in a

different way and in one that I would challenge which is very intimidating to be honest um but in this in this essay on symbolism he talks about essentially two types of symbolism the symbolism of the church and the symbolism of language and of algebra so he initially says that there is a symbolism of the medieval cathedral but because that symbolism can be dispensed with it is on The Fringe of life and has an unessential element in its Constitution and then he says and on page two here there is another sort of language a written language which is constituted by the mathematical symbols of the science algebra in some ways these symbols are different to those of ordinary language because the manipulation of the algebraic symbols does your reasoning for you provided that you keep to the out algebraic rules how can something do reasoning for you if it is purely mechanistic and has no agency and then he says this is not the case with ordinary language which puts up that boundary again between words and symbols that the symbols can do something the words cannot so I would disagree here with how he draws the bifurcation between Cathedrals and mathematical symbols uh Pythagoras would also have quibbled with that pythagora for the pythagoreans mathematics was very much a religious practice and they believed that numbers had a spiritual Dimension but that takes us to a very weird that takes us to very weird places very quickly uh I tweeted this I don't necessarily believe it but it's you don't have to tweet everything you believe or vice versa uh so let's go to slightly a bit more out there but this is coming back to relevance I promise it has to do with the notion of sovereignty and so here I'm looking at the article sovereignty in the UFO by win and Duvall where they talk about how sovereignty is constructed and so this might go back to our definition here of a quote unquote real world an institution a government a nation state even an individual as being Sovereign has a particular Construction in the world of our experience but this has not always been the case I loved this quote that they said anthropos anthropocentric sovereignty might seem necessary after all who else besides humans might rule nevertheless sovereignty was LE historically sovereignty was less anthropocentric for Millennia nature and the gods which nature would be in science and the Gods be and religion were thought to have causal powers and subjectivities that enable them to share sovereignty with humans if not exercise Dominion outright authoritative belief in non-human sovereignties was given up only after long and bitter struggle about the borders of the social world so again we're trying to place a border in which who what could be Sovereign depends on who what should be included in society and there's a great footnote here anthropocentrism need not mean all human beings since historically many physical humans were not considered human socially and let me go back here and say that perhaps there's a

similar construction occurring perhaps this division some humans are human and some are not some we would not recognize that today but we do recognize bation in the math persons and non-math persons and I would say it's equally as social and equally as artificial and equally as dangerous so where are we now U so I've opened some doors here perhaps some of them should have stayed shut but regardless they're open for the moment and we can shut them when we wish hopefully I love this quote at the end about sovereignty in the UFO the UFO can be known only by not asking what it is uh because this has been totally obverse to my process here I have tried to know mathematics by asking what it is by struggling very hard with its identity as an egregor as a concept um but maybe some things can only be known by not asking what they are so I'm sure that this has perhaps been an interesting metaphysical IAL exploration of some different historical ideas and ideas of History how is this applicable to ourselves as scholars in an active inference environment well what I would like to point at is the ways in which some of these ideas that are already Dynamic may be changing very rapidly at this moment so if we look at the identity of mathematician if we are solely confined to a meatspace identity that has certain connotations but what do we do with someone like bluecow and I did request blue Cal's consent to use their likeness and their thoughts uh the creator of the first hyperdimensional Computing OS and the super prompt a prompt God uh someone who is a self-proclaimed mathematician who does impressive mathematical work in prompting in prompt engineering of mathematical prompts via GitHub and whose appear par is a an anime person dressed as a cow um can they be a real mathematician in the way that we currently conceptualize math and are the mathematical ideas that they're bringing ones that we are prepared to reckon with as math and then when they post things like this what do we do self-identified skitso and it's helpful here perhaps to remember that skitso means split another scholar in this emergent might one might call parallel or underground genre whose work I respect is Sir Loop and Watson his conjecture soon you'll be able to buy an AI hacker for less than \$100,000 he'll or she do he he he'll or or she do anything for free you'll need to pay only once and it will last forever this is exactly the bargain of the theorem once you have proven a theorem once with a finite amount of human effort it remains irrevocably true and will do whatever you ask it to do for free forever so if we believe there is a world where theorems come from what does that say about other objects or other constructions that might share their properties I don't have a conclusion if I were going to have a conclusion it would be it going back to the thoughts of slide three but one reason I don't have a conclusion is that this is not the end of a completed research Journey but the beginning of something we are all

experiencing together so I'm gesturing at what I know and then the next question is what do I hope to know well one challenge that I think this immediately offers but also an opportunity is what does it look like to be an institution in a world where these agents these human agents are seriously considering the same ideas as you are and perhaps offering insights that are not being produced at the level of your institution I don't this may be Fringe but I think it's to be taken seriously and I offer it because many people will see the work that for instance Terry tow is doing on leading larger mathematical teams that use AI to modularize math and prove more efficiently in an engineering type framework different from the idea of the individual math um it's less likely that the average academic I believe would encounter these types of people who are also doing interesting mathematical thinking not only about the mathematics itself but about what it means to present oneself as a mathematician or to present mathematical work so I hope these thoughts have been interesting to you and excited to see what they may have brought up in the form of questions and comments thank you ever so much octopus um I really appreciate the number of different ways that you've kind of viewed this topic um if I may start with with a question sure so what as you were talking about this idea that the formulas maybe exist and that they're kind of waiting to fit on to something it was reminding me of a um a song by the Icelandic uh artist Bor I'm not sure if you're familiar with her she has a a song called modern things where it says all those modern things they've always existed they've just been waiting in a mounting for the right moment um and it's their time now and um so yeah I was wondering you know what what do you think about um something like the global economy which it's um it kind of acts like an entity in itself so it's you know it's the Invisible Hand of the market for example but it's the humans that kind of assemble around it and imbue the meaning on the all the uh interactions and all Dynamics yes uh I think again that there again there might be a border uh for instance if we view economics as purely human transactions of material Capital it misses a lot I think a lot of economists would say that but symbols are important even there if you look at a unit of currency it is imbued with symbols that are highly non-functional almost overridden with nonfunctional and one of the ways that one can assert a Authority and or immortality is to issue a coin and that's what we see now with cryptoeconomic systems is a debate around the individual sovereignty and the right of who is allowed to issue something that can be used as a unit of exchange uh but that's not a new debate there were certain Roman citizens who had the authority to issue a token and then the question is uh where does the value of that token come from ultimately right so so again it kicks back very quickly to questions around uh beliefs and identity and sovereignty I guess the

thing there is when something is real to me it must have more than just a material component
thank you um Daniel are there any questions from the chat there's a lot of great commentary
octopus it was a great presentation do you have any final comments oh um well I really
appreciate the invitation to be heard and the ability to show up as octopus that's so meaningful
to me and U grateful to the organizers and the audience and looking forward to seeing what
conversations can emerge from this beginning awesome thank you till next time see you ooto
thank you for